

- 2 Domestic goats are small, herbivorous animals that provide milk for human use. This is an important source of food for people in rural South Africa.

Three Northern European goat varieties (breeds) have been imported to South Africa because they have higher milk yields than the native South African goats.

Table 2.1 compares the mean daily milk yields of these three breeds of Northern European goat in three locations, Northern Europe, Barbados and South Africa.

**Table 2.1**

| goat breed     | mean daily milk yield/kg<br>in different locations |          |              |
|----------------|----------------------------------------------------|----------|--------------|
|                | Northern Europe                                    | Barbados | South Africa |
| British Alpine | 4.09                                               | 2.55     | 0.75         |
| Saanen         | 5.17                                               | 1.73     | 1.45         |
| Toggenburg     | 4.54                                               | 3.46     | 0.56         |

- (a) Native South African goats have a mean daily milk yield of 0.25 kg.

Calculate how many times greater the mean daily milk yield will be if a native South African goat is replaced by the Northern European goat breed that gives maximum yield.

Show your working and write your answer to **two** significant figures.

answer ..... [2]

- (b) (i) Explain how the data in Table 2.1 support the claim that some of the variation in mean daily milk yield in goats is due to **genetic** causes.

.....

.....

.....

.....

.....

..... [2]

- (ii) The climate, vegetation and availability of veterinary care for goats in Northern Europe, Barbados and South Africa are different.

Explain how Table 2.1 shows that **environmental** factors can cause variation in mean daily milk yield in goats.

[2]

- (c) Native South African goats are better adapted to the local conditions in South Africa than a Northern European breed, such as the Saanen. However Saanen goats have the potential for very high milk yield.

Outline a programme of selective breeding that could produce a goat with a high milk yield that is adapted to the local conditions in South Africa.

[5]

- (d) Children in developing countries may drink unpasteurised goats' milk. Some may develop diarrhoea caused by live bacteria ingested in the milk.

Scientists have used genetic engineering to develop goats that produce human lysozyme in their milk. Lysozyme is an enzyme that kills bacteria and so reduces the number of bacteria in the milk.

State a social advantage **and** a social disadvantage of making these GM goats available in developing countries.

advantage .....

.....

disadvantage .....

.....

[2]

[Total: 13]